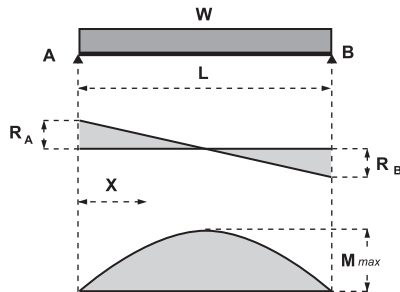


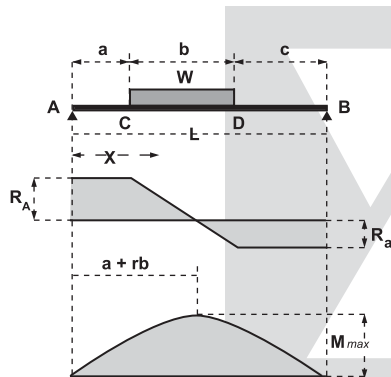
Simply supported beam



Uniform load on full span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total uniform load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL^2}{8} \\ \delta_{\max} &= \frac{5}{384} \cdot \frac{WL^3}{EI} \end{aligned} \right. \\
 i_A = i_B &= \frac{WL^2}{24EI} \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{2L}(L-X) \\ \delta_x &= \frac{WX}{24EI}(X^3-2X^2L+L^3) \\ i_x &= \frac{W}{24EI}(4X^3-6X^2L+L^3) \end{aligned} \right.
 \end{aligned}$$

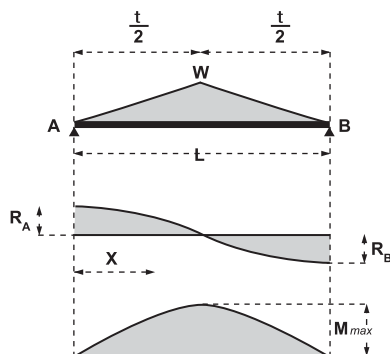
Simply supported beam



Uniform load on part of span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total uniform load} &= W \\
 \text{Let } r &= \frac{0.5b+c}{L} \\
 R_A = Wr; \quad R_B &= W(l-r) \\
 \text{at } X = a + rb, M_{\max} &= Wr(a+0.5rb) \\
 i_A = \frac{Wr(L^2-c^2-Lbr)}{6EI}; \quad i_B &= \frac{W(l-r)}{6EI}[L^2-a^2-Lb(l-r)] \\
 \text{Equation to elastic line between C and D, i.e. } a \leq X \leq a+b & \\
 \delta_x &= \frac{W}{24EIb}[X^4-4(a+rb)X^3+6a^2X^2+4\{rb(L^2-c^2-cb-\frac{b^2}{2})-a^2\}X+a^4]
 \end{aligned}$$

Simply supported beam



Triangular load on full span

$$\begin{aligned}
 \text{Span} &= L \\
 \text{Total load} &= W \\
 R_A = R_B &= \frac{W}{2} \\
 \text{at mid-span} &\left\{ \begin{aligned} M_{\max} &= \frac{WL^2}{6} \\ \delta_{\max} &= \frac{WL^3}{60EI} \end{aligned} \right. \\
 i_A = i_B &= \frac{5WL^2}{96EI} \\
 \text{at } X \text{ from A} &\left\{ \begin{aligned} M_x &= \frac{WX}{6L^2}(3L^2-4X^2) \\ \delta_x &= \frac{WX}{480EI}(16X^4-40X^2L^2+25L^4) \\ i_x &= \frac{W}{96EI}(16X^4-24X^2L^2+5L^4) \end{aligned} \right.
 \end{aligned}$$